

Figure 1-10: Phase diagrams and microstructures of the Pb-Sn system.

Phase Diagrams (a, b, c, d): These diagrams show the temperature-composition relationship for the Pb-Sn system. The x-axis represents the composition in wt.% Sn (0 to 100), and the y-axis represents the temperature in °C (1000 to 1500). The diagrams illustrate the cooling of different alloys and the resulting phase transformations.

- (a) Cooling of 38.1% Pb-61.9% Sn alloy:** Shows the formation of primary eutectic (L + S₂) and eutectic (L + S₁) phases.
- (b) Cooling of 60% Pb-40% Sn alloy:** Shows the formation of primary solid solution (L + S₂) and eutectic (L + S₁) phases.
- (c) Cooling of 60% Pb-40% Sn alloy:** Shows the formation of primary solid solution (L + S₂) and eutectic (L + S₁) phases.
- (d) Cooling of 60% Pb-40% Sn alloy:** Shows the formation of primary solid solution (L + S₂) and eutectic (L + S₁) phases.

Phase Compositions and Microstructures:

Temperature (°C)	Phases	Composition	Mass (%)	Microstructure and Comment
250	L	40% Sn	100	The first solid (α-phase) nucleates in the liquid.
235	L	40% Sn	100	
210	α	15% Sn	0	
210	L	50% Sn	68.7	Mixture of liquid and α phases. More solid forms. Compositions change.
183.5	L	50% Sn	31.3	Liquid has the eutectic composition.
183.5	L	19.2% Sn	51.3	
182.5	L	19.2% Sn	73.7	Eutectic (α and β phases) and primary α-phase.
182.5	L	97.5% Sn	26.6	

Eutectic Reaction and Microstructure:

The eutectic reaction is represented by the equation: $L \rightarrow S_1 + S_2$. The microstructure shows the formation of the eutectic structure, which consists of alternating layers of the two solid phases (S₁ and S₂).

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